

Frictions in DeFi Liquidations: Evidence from the Aave V2 Main Market

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Abstract: Lending in decentralized finance (DeFi) relies on collateral and efficient liquidations to manage credit risk. The permissionless and pseudonymous nature of public blockchains precludes reputation-based lending in DeFi and renders liabilities effectively non-recourse. Frictions in collateral liquidations increase the risk of bad debt and may ultimately lead to protocol defaults and losses for liquidity providers. This paper studies liquidation dynamics in the Aave V2 Main Market on Ethereum using block-level data covering 46 months and more than 54 000 borrower positions. While most undercollateralized debt is liquidated almost instantaneously, a non-trivial share of positions remains open for extended periods. Using a state model to distinguish healthy, viable for liquidation, and stale borrower positions, this paper quantifies transition probabilities and identifies factors associated with liquidation success. Logistic regression results show that liquidation size, lower network transaction fees, and relative profitability are associated with the probability of liquidation success in the subsequent block. At the same time, oracle price distortions and asset price volatility are associated with lower liquidation likelihood, consistent with heightened execution risk. The findings provide new high-frequency evidence on liquidation frictions in a large and mature DeFi lending market. The results contribute to the understanding of the microstructure of DeFi liquidations and credit risk in decentralized lending protocols.

Keywords: Liquidations, Lending Markets, Decentralized Finance, Credit Risk, Blockchain.

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1 Introduction

Collateral has a long history in financial contracting as a means of mitigating credit risk and aligning borrower and lender incentives. It reduces information asymmetry and enhances confidence in extending credit by pledging assets against liabilities. Its effectiveness, however, depends critically on the ability to timely liquidate in case borrowers fail to meet their obligations. Understanding liquidation dynamics is therefore essential for assessing risk in collateral-based lending markets.

In the context of blockchain-based permissionless financial infrastructure or decentralized finance (DeFi), collateral takes a central role in any form of lending. Public blockchains pose a pseudonymous environment, where identities in the form of addresses can be switched at negligible cost. This precludes reputation-based lending and renders liabilities effectively non-recourse. Collateral constitutes the exclusive form of credit risk mitigation and efficient liquidations are central to protecting DeFi applications, commonly referred to as protocols, and their liquidity providers from bad debt and subsequent losses.

Lending constitutes one of the largest protocol categories in DeFi. At the time of writing, it accounts for more than half of the USD 167 billion in total value locked across DeFi ecosystems.³ Despite their scale, lending protocols operate without discretionary intervention. Smart-contract rules, protocol parameters, oracle prices, transaction costs, and incentives offered to external liquidators determine liquidation outcomes. Frictions at any of these layers may delay or prevent liquidations, increasing the likelihood of bad debt. Accordingly, the notion of liquidation efficiency used in this paper refers to the protocol’s interest in timely risk resolution, not to the optimization problem of liquidators.

As the recent incident around rsETH and its effect on the Aave V3 lending market illustrates, the economic relevance of credit risk for liquidity providers is not merely theoretical. While the originating failure was outside Aave V3 itself and the event differs from liquidation frictions studied in this paper, it underscores that bad debt can become a direct balance sheet concern.⁴

A growing literature at the intersection of economics and computer science studies DeFi lending with regard to market design, liquidation mechanisms, as well as feedback loops with asset prices and market liquidity. Although prior work emphasizes the relevance of liquidations, empirical evidence on frictions is limited. Existing studies on positions failing to liquidate often rely on coarse time aggregation, leaving micro-level dynamics underexplored.

This paper contributes a high-frequency empirical analysis of liquidation frictions in decentralized lending markets. It studies how positions evolve and what factors are associated with liquidation outcomes at block level, using data from the Aave V2 Main Market on Ethereum

³Data from DeFiLlama.com, retrieved on October 4, 2025.

⁴See, for example, Aave’s incident report on April 20, 2026: <https://governance.aave.com/t/rsETH-incident-report-april-20-2026/24580>

(Aave V2). The observations cover 46 months and 54 127 borrower positions in one of the largest and most mature DeFi lending protocols.

To structure the analysis, the paper proposes a state-based framework that classifies borrower positions as healthy, viable for liquidation, or stale, considering protocol risk parameters as well as transaction cost of liquidations. The analysis of transition probabilities between these states shows that once a position is not liquidated immediately, its collateralization level is more likely restored through market changes, such as oracle price updates.

Focusing on positions viable for liquidation, the paper estimates the probability that a liquidation occurs in the subsequent block. Logistic regression results indicate that liquidation size, relative profitability, and low network fees are strongly associated with liquidation success. At the same time, oracle price distortions and asset price volatility are associated with reduced liquidation probability.

The findings contribute new empirical evidence on the microstructure of DeFi liquidations and highlight mechanisms through which liquidation frictions can arise even in mature lending markets. The presented results may provide useful insights to practitioners and inform further research on DeFi lending market design and risk management.

The remainder of the paper is structured as follows. Section 2 introduces DeFi lending markets in general, relevant Aave V2 specifics, and shares an overview of related literature. Section 3 proposes a Markov-like state model for borrowing positions and describes the data collection. Section 4 introduces the empirical methodology and presents the analysis results. The robustness of findings is examined in Section 5, covering subsample analyses and endogeneity concerns. The main findings are summarized in Section 6.

2 Background

This section introduces the conceptual background for this study. It outlines the general mechanics of DeFi lending markets, specifics of liquidations in Aave V2, and situates the paper within the related literature. A short introduction to fundamentals of liquidations in DeFi can be found in Appendix A.

2.1 Lending Markets Primer

Fundamentally, DeFi lending markets are smart contracts deployed on a public blockchain network that form an open infrastructure. Anyone can provide liquidity in the market’s reserve assets and either simply earn interest (savers) or use their deposits as collateral to borrow from the market liquidity (borrowers).

Lending-market contracts encode the rules and parameters governing liquidity provision and borrowing across a defined set of reserve assets. They hold market liquidity and track

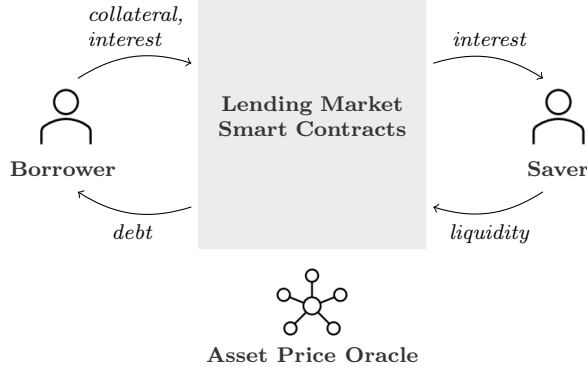


Figure 1: Main components of a DeFi lending market distinguishing the two user categories of savers, who provide assets as liquidity to the market to earn interest, and borrowers, who use the assets they provide as collateral to borrow from the market liquidity. The lending market smart contracts entail the rules and parameters for lending and borrowing across reserve assets and hold market liquidity. The oracle is the source of current asset prices for the lending market to continuously track user balances.

the state of all active positions, including deposit and borrow balances with accrued interest. Usage-dependent interest rate models manage liquidity and maturity transformation. A central component is the price oracle, which supplies current reserve asset prices. Using these oracle prices, the market determines borrowing capacity, monitors collateralization, and prices assets during liquidation.

Formally, consider a DeFi lending market with n reserve assets and m user positions, indexed by their unique blockchain address k . With $i \in \{1, \dots, n\}$, each position is characterized by its current deposit $d_i^{(k)}$ and borrow $b_i^{(k)}$ balances. Based on the oracle prices p_i , the market tracks the total value of a position's deposits and borrows at any given time:

$$D^{(k)} = \sum_{i=1}^n (d_i^{(k)} \cdot p_i)$$

$$B^{(k)} = \sum_{i=1}^n (b_i^{(k)} \cdot p_i).$$

To determine borrowing capacity and monitor credit risk, the market assigns each reserve asset a maximum loan-to-value parameter $\bar{\varphi}_i$. A position is considered *healthy*, as long as its outstanding borrows do not exceed the borrowing capacity defined by its deposits that serve as collateral. This is captured by the health factor

$$h^{(k)} = \frac{\sum_{i=1}^n (d_i^{(k)} \cdot p_i \cdot \bar{\varphi}_i)}{B^{(k)}}, \quad (1)$$

s. t. $B^{(k)} > 0$,

Positions with $h^{(k)} < 1$ are *unhealthy* and in violation of the lending market's credit risk conditions. While unhealthy, positions are open for liquidation.

Lending markets generally employ fixed-rate, permissionless liquidation mechanisms. When a position is unhealthy, liquidators can repay (part of) its debt and receive (part of) the position's deposits serving as collateral at a discounted rate. Consider a liquidation where the liquidator repays k 's debt in reserve asset i and receives collateral in reserve asset j . The protocol incentivizes liquidators by offering a bonus ϕ_j , so the liquidator receives collateral at a factor of $(1 + \phi_j)$ times the value of repaid debt v . In practice, liquidators face execution risk due to potential deviations between oracle prices p and market prices $\dot{p}$. Liquidators also incur transaction costs τ , including blockchain network fees. Accounting for these factors, liquidator profit can be expressed as:

$$r_{i,j}^{(k)} = v \cdot \left(\frac{\dot{p}_j \cdot (1 + \phi_j)}{p_j} - \frac{\dot{p}_i}{p_i} \right) - \tau, \quad (2)$$

s.t. $v \leq \min \left(\frac{d_j^{(k)} \cdot p_j}{1 + \phi_j}, b_i^{(k)} \cdot p_i \right),$

$h^{(k)} < 1.$

During liquidation, the liquidatable amount v is bounded by the position's current balances of both debt and collateral assets. Specifically, v cannot exceed the smaller of the position's debt in asset i and the position's collateral in asset j adjusted for the liquidation bonus ϕ_j . Notably, transaction costs include a fixed component due to blockchain network fees while liquidation incentive ϕ_j scales with liquidation volume.

Efficient liquidations are central to managing credit risk in DeFi lending markets. In the following, efficiency is defined from the perspective of a lending market rather than from that of liquidators. A liquidation is efficient when an insufficiently collateralized position is liquidated immediately, thereby limiting the protocol's risk of incurring bad debt. This differs from the liquidator perspective, where delayed execution may be optimal if expected profits are low relative to execution risk. The paper focuses on the former notion, treating frictions as delays or failures in resolving risky positions from the perspective of protocol solvency and hence protection of liquidity providers. Figure 2 illustrates how write-offs to user debt may be absorbed by protocol reserves or, ultimately, affect liquidity providers.

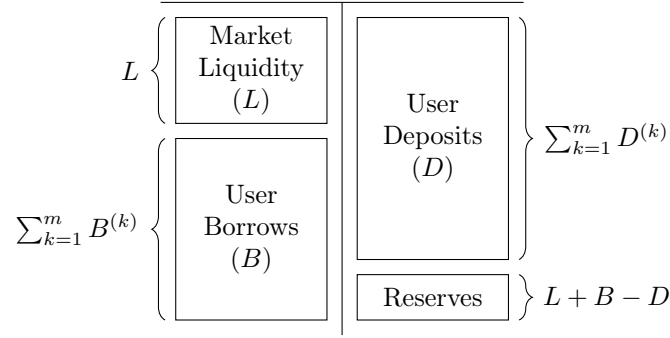


Figure 2: Stylized balance sheet of a lending market. The asset side entails the market liquidity (L), i.e., liquidity across reserve assets available for borrowing, withdrawal, or liquidation, and the total amount borrowed by users, including accrued interest (B). The liability comprises the total user deposits including accrued interest (D) as well as the protocol reserves as the residual of total assets after liquidity provision claims.

2.2 Aave V2 Main Market

The empirical analysis focuses on Aave V2, one of the largest DeFi lending protocols during the observation period. Two idiosyncrasies are particularly relevant for liquidation dynamics. First, borrowers can designate whether a deposited asset is used as collateral. This is captured by a binary indicator $a_i^{(k)}$, where $a_i^{(k)} = 1$ indicates the use of the deposit balance in i as collateral. Consequently, the health factor introduced in Equation 1 is modified to include only deposits designated for collateral use:

$$h^{(k)} = \frac{\sum_{i=1}^n (d_i^{(k)} \cdot a_i^{(k)} \cdot p_i \cdot \bar{\varphi}_i)}{B^{(k)}}, \quad (3)$$

s. t. $B^{(k)} > 0$.

Second, Aave V2 limits the amount that can be liquidated in a single transaction through a close factor $CF = (0, 1]$. This parameter caps the maximum liquidatable debt at a fraction of the position's total borrow balance $B^{(k)}$. While this design protects borrowers from having their entire position liquidated at once, it also means that positions cannot generally be fully closed through a single liquidation transaction. Equation 2 is therefore amended as follows:

$$r_{i,j}^{(k)} = v \cdot \left(\frac{\dot{p}_j \cdot (1 + \phi_j)}{p_j} - \frac{\dot{p}_i}{p_i} \right) - \tau, \quad (4)$$

s.t. $v \leq \min \left(\frac{d_j^{(k)} \cdot a_j^{(k)} \cdot p_j}{1 + \phi_j}, CF \cdot B^{(k)}, b_i^{(k)} \cdot p_i \right),$

$h^{(k)} < 1$.

The close factor interacts with liquidation incentives in an important way. Transaction costs τ include a fixed component in the form of blockchain network fees while the liquidation

bonus ϕ_j scales with volume v . As partial liquidations shrink positions, remaining liquidation opportunities may fall below profitability thresholds. This mechanism can give rise to small positions, also referred to as dust positions, that remain unhealthy for extended periods despite being mechanically eligible for liquidation.

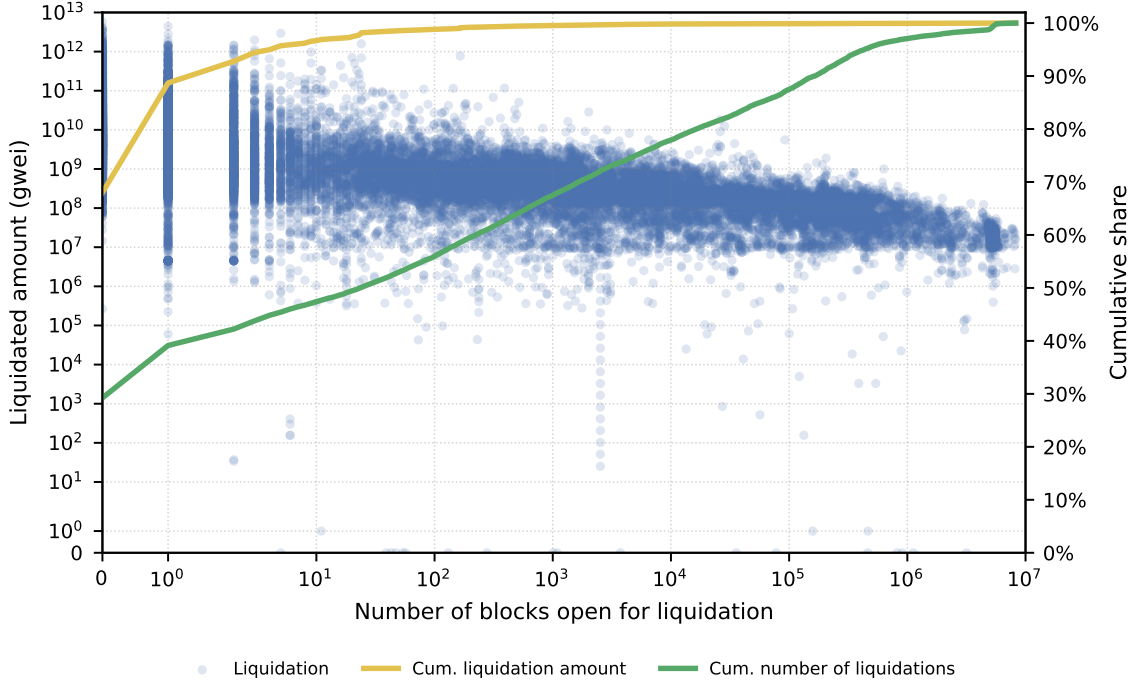


Figure 3: Overview of the 33 168 liquidation events in the Aave V2 Main Market on the Ethereum mainnet between December 2020–September 2024 in terms of the debt liquidated in gwei (10^{-9} Ether) and the duration for which the position has been unhealthy prior to being liquidated. Each dot represents one liquidation event.

The intuition of a lower bound in terms of volume of successful liquidations is consistent with the pattern discernible in Figure 3. Each blue dot represents a successful liquidation event in Aave V2, categorized by volume in terms of debt repaid and consecutive blocks the position has been unhealthy before being liquidated.

The pattern in this initial descriptive analysis suggests a positive correlation between liquidation size and exploitation speed. Overall, 68.0% of the total 621 036 Ether in liquidated debt are picked up in the same block the position became unhealthy. This share rises to 95.9% within 5 blocks which equals approximately 1 minute. However, there is a notable difference when looking at the cumulative number of liquidations in relation to their exploitation delay. The results indicate that 49.7% of liquidated positions have been unhealthy for more than 25 blocks or approximately five minutes. 23.5% exceed the 1-day and 8.4% the 30-day mark of 7200 and 21 600 blocks.

While these initial statistics ignore potential unhealthy positions that are never liquidated,

they convey that even with successful liquidations, there appear to be frictions causing delay and thus exacerbating credit risk. This motivates a closer examination at position level.

2.3 Related Work

The academic literature on DeFi lending covers various aspects of protocol design, user behavior, and risk dynamics. Early work studies lending market mechanics and identifies vulnerabilities and attack vectors (e.g., Gudgeon et al., 2020b; Qin et al., 2021; Bartoletti et al., 2021). Subsequent research analyzes borrowing behavior, leverage, and cross-market interdependencies, highlighting pro-cyclical dynamics and potential contagion channels (Darlin et al., 2022; Saengchote, 2023; Heimbach and Huang, 2024; Moallemi and Patange, 2024; Aspris and Svec, 2026; Cornelli et al., 2025; Gregory and Vito, 2025).

A prominent strand of research focuses on liquidations and their interaction with asset prices and market liquidity. Several studies analyze the feedback loop between liquidations and price dynamics in decentralized exchanges, emphasizing the potential for systemic fragility during periods of stress (Lehar and Parlour, 2023; Chiu et al., 2023; Sasi-Brodesky and Nassr, 2023). Nadler et al. (2026) study the role of asset price oracle accuracy in the context of capital efficiency. Other work adapts established risk and contagion frameworks to the DeFi lending context (Gudgeon et al., 2020a; Schuler et al., 2023; Tovanich et al., 2025). While these studies highlight the potential for liquidations to amplify market stress, they focus primarily on aggregate dynamics.

Empirical evidence on micro-level frictions in DeFi liquidations is relatively scarce. Perez et al. (2021) document a marked reduction in delayed liquidations during the early days of DeFi lending. More closely related to this paper, Campello et al. (2023) and Mu et al. (2025) study open or near-liquidation positions based on daily snapshot data. This paper extends the literature by providing a high-frequency, block-level analysis of factors associated with liquidation success of unhealthy positions in a major DeFi lending market.

3 State Model and Data Description

This section introduces a Markov-like state model that characterizes borrower positions in DeFi lending markets at block frequency. The model formalizes liquidation eligibility and economic viability and categorizes state transitions. The section then describes the block-level dataset underlying the empirical analysis.

3.1 Position States and Transition Causes

To study liquidation dynamics at high-frequency, borrower positions are classified into discrete states based on mechanical liquidation eligibility and the economic viability of liquidation,

reflecting protocol rules and transaction cost considerations.

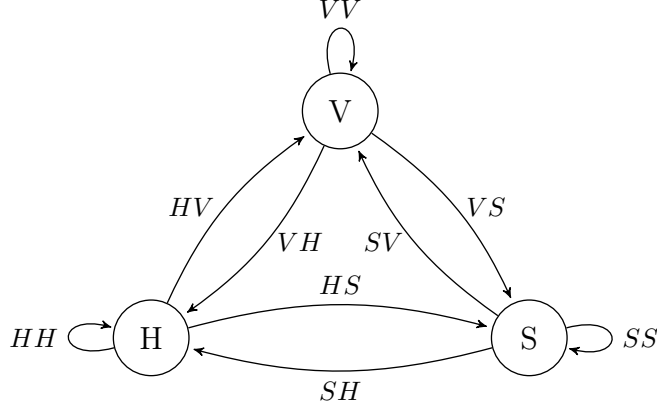


Figure 4: A Markov-like state model for borrower positions. In the *healthy* state (H), the position is sufficiently collateralized ($h^{(k)} \geq 1$) and mechanically excluded from liquidation. The *viable* state (V) comprises positions that are open for liquidation due to violation of protocol risk parameters and offer sufficient liquidation volume to be economically profitable ($h^{(k)} < 1 \wedge \bar{v}^{(k)} \geq \theta$). The *stale* state S applies to borrower positions that are open for liquidation but their volume is too small in light of transaction costs to be profitable ($h^{(k)} < 1 \wedge \bar{v}^{(k)} < \theta$).

Borrower positions are healthy (H) whenever $h^{(k)} \geq 1$. Positions in (H) satisfy protocol risk parameters and are protected from liquidation. In case the health condition is not met, protocol rules classify the position as undercollateralized and permit its liquidation. For the purpose of this study, unhealthy positions are further divided into viable (V) and stale (S) states, depending on whether liquidation is economically attractive. Specifically, an unhealthy position is classified as viable for liquidation (V) when its maximum liquidatable debt volume is sufficiently large for proceeds to cover transaction cost. Conversely, an unhealthy position is classified as stale (S) when the liquidatable amount is too low to constitute an economically viable liquidation opportunity. To operationalize this classification, the analysis introduces a threshold θ in the form of a minimum amount of debt that can be liquidated in a single transaction. Unhealthy positions with maximum liquidation volume $\bar{v}^{(k)} \geq \theta$ are classified as viable (V), while those with $\bar{v}^{(k)} < \theta$ are deemed stale (S). This threshold should be interpreted as proxy for minimum economically viable liquidation size rather than a behavioral cutoff. Misclassifications are to be expected, and alternative values for θ will be considered as part of robustness checks.

Figure 4 summarizes the borrowing position states and possible transition paths from one period to the next. Generally, a position can transition between any state on a direct path. However, if we differentiate between transition causes, there are rule- and incentive-based limitations, summarized in Figure 5.

The first category of transition causes entails any change to market conditions such as interest accruals, oracle price changes, or risk parameter updates. These changes can affect both

	HH	HV	HS	VV	VH	VS	SS	SH	SV
Δ Market Conditions									
Borrower Action									
Liquidation									

Prevented by protocol logic

No incentive

Possible and (maybe) incentivized

Figure 5: Overview of state transition paths highlighting rule and incentive-based restrictions by cause. Market condition changes include any parameter update affecting how the protocol values the position’s debt relative to the deposited collateral or what the maximum liquidation volume is. Borrower actions refer to any changes to deposited collateral and drawn debt by the user behind a position. Liquidation comprises all changes to a position through liquidators.

the health factor and the maximum liquidation volume in either direction and can therefore cause any state transition. The second category of borrower actions entails any changes to the asset balances deposited or borrowed by the user controlling the position. As protocol rules prevent actions that render a position unhealthy, transitions out of (H) cannot be caused by the borrower. Once in the unhealthy states (V) or (S), protocol rules also prohibit any borrower intervention that further reduces position health. Accordingly, the borrower action set is restricted to either repaying debt or depositing additional collateral. A borrower’s only primary incentive to do so is to unlock the position’s collateral or to prevent liquidation. In consequence, only borrower actions that reinstate position health are rational. Finally, the restrictions for transitions caused by liquidations are intrinsic to the state. Positions in (H) are mechanically blocked from liquidation while positions in (S) typically offer insufficient profitability. Starting from (V), a liquidation can in principle result in any subsequent state. A liquidation does not necessarily reinstate a position’s health or reduces its size sufficiently for it to be considered stale.

3.2 Data

The empirical analysis uses a novel block-level dataset for the Aave V2 Main Market deployed on Ethereum. The sample spans from the launch of Aave V2 on November 30, 2020 to September 30, 2024, covering 9 504 357 blocks. It entails all 54 127 positions that showed a positive borrow balance at the end of at least one block. Borrower positions are tracked continuously from the first protocol interaction until they are fully closed or the sample ends. Focusing on a single mature lending protocol allows for a consistent mapping of protocol rules and liquidation mechanics over time.

Based on the analysis of the source code of deployed Aave V2 smart contracts, the dataset is constructed by parsing on-chain events, including deposits, borrows, repayments, withdrawals, and liquidations. These events are combined with reserve asset configuration parameters, oracle

prices, and Ethereum block metadata, all sourced using public APIs for Ethereum archive nodes by *Infura* and *Alchemy*. Figure 6 summarizes key time-series aggregates. A preliminary inspection of the data yields similar conclusions as the initial view on liquidation efficiency in Figure 3. Although unhealthy debt never exceeds 0.25% of total debt measured at the end of each day, it persists throughout the observation period, indicating frictions in liquidations warranting further analysis.

In a next step, liquidation thresholds, health factors and maximum liquidation volume are reconstructed at block frequency for each borrower position, yielding observations at position-block pair level. These observations are then classified into the states as defined in Section 3.1 and form the basis for the transition probability analysis in Section 4.1.

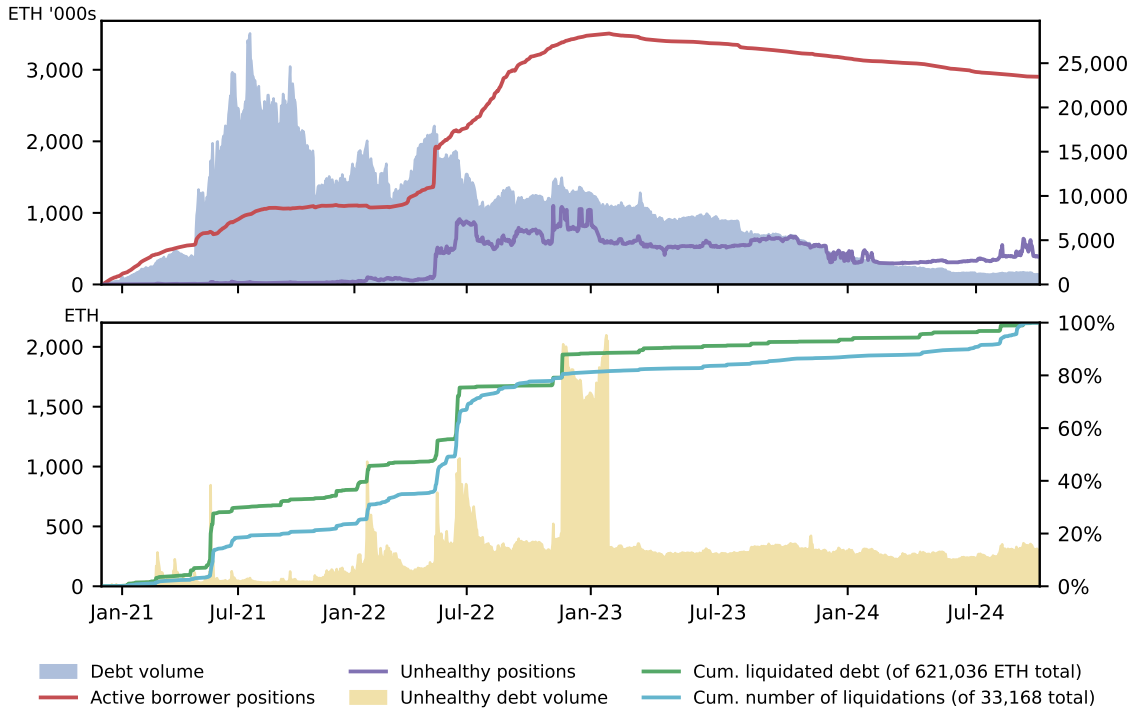


Figure 6: Overview of the Aave V2 Main Market on Ethereum showing the evolution of total debt, unhealthy debt, number of open borrower positions both healthy and unhealthy, as well as liquidations in terms of volume and events for November 30, 2020 – September 30, 2024. For display purposes, collected data are aggregated to daily snapshots.

The main analysis in Section 4.2 focuses on parameters associated with liquidation probability of positions in (V), using a baseline threshold of $\theta = 250$ finney, equivalent to 0.25 Ether. The data on position state in each block is complemented with variables capturing transaction cost and price risk inherent to the expected liquidation, as well as block building characteristics. Table 2 presents summary statistics for the independent variables used in the liquidation probability analysis for viable positions with a minimum liquidation volume of 250 fin. A detailed description of the variables and their construction logic can be found in Appendix B.

Table 1: Reserve asset overview of Aave V2 Main Market on Ethereum November 2020 – September 2024. Alphabetic list showing the month of initialization and the role over time. As long as an asset is active for variable or stable lending, users can increase their borrowing amount in these assets against suitable collateralization. Phases of collateral use are defined by the asset having a LTV of $\geq 20\%$ and new deposits being accepted. The phase out is defined by the reserve factor, i.e., the proportion of interest paid by borrowers retained by the protocol instead of credited to the liquidity providers, raising to $\geq 90\%$, effectively taking away the incentive for liquidity provision in the asset.

Symbol	Name	Initialized	Variable Lending	Stable Lending	Collateral Use	Phase-Out
1INCH	1inch Token	Jul 22	Jul 22–Nov 22		Jul 22–Jun 23	Dec 23
AAVE	Aave Token	Dec 20	Dec 20–Dec 20		Dec 20–Apr 24	
AMPL*	Ampleforth	Jul 21	Jul 21–Nov 22			Dec 23
BAL	Balancer	Feb 21	Feb 21–Oct 22		Feb 21–Nov 22	Dec 23
BAT	Basic Attention Token	Dec 20	Dec 20–Oct 22	Dec 20–Oct 22	Dec 20–Nov 22	Dec 23
BUSD	Binance USD	Dec 20	Dec 20–Feb 23			Mar 23
CRV	Curve DAO	Dec 20	Dec 20–Nov 22		Dec 20–Aug 23	Apr 24
CVX	Convex Finance	Jun 22	Jun 22–Oct 22		Jun 22–Nov 22	Dec 23
DAI	Dai Stablecoin	Dec 20	Dec 20–	Dec 20–Aug 23	Dec 20–	
DPI*	DefiPulse Index	Aug 21	Aug 21–Nov 21		Aug 21–Mar 23	Dec 23
			Jun 22–Oct 22			
ENJ	Enjin Coin	Dec 20	Dec 20–Nov 22	Dec 20–Nov 22	Dec 20–Nov 22	Dec 23
ENS	Ethereum Name Service	Mar 22	Mar 22–Nov 22		Mar 22–Jun 23	Feb 24
FEI*	Fei USD	Sep 21	Sep 21–Sep 22		Jan 22–Sep 22	Sep 22
FRAX*	Frax Dollar	Sep 21	Sep 21–Apr 24			May 24
GUSD*	Gemini Dollar	Jan 21	Jan 21–Apr 24			May 24
KNC	Kyber Network Crystal	Dec 20	Dec 20–Apr 22	Dec 20–Apr 22	Dec 20–Apr 22	Dec 23
LINK	ChainLink Token	Dec 20	Dec 20–Nov 22	Dec 20–Nov 22	Dec 20–Jun 23	
LUSD*	Liquid USD	Aug 22	Aug 22–Apr 24	Aug 22–Nov 23		May 24
MANA	Decentraland	Dec 20	Dec 20–Nov 22	Dec 20–Nov 22	Dec 20–Nov 22	Dec 23
MKR	Maker DAO	Dec 20	Dec 20–Nov 22	Dec 20–Nov 22	Dec 20–Jun 23	Apr 24
RAI*	Rai Reflex Index	Jun 21	Jun 21–Nov 22			Dec 23
REN	Republic Token	Dec 20	Dec 20–Oct 22	Dec 20–Oct 22	Dec 20–Nov 22	Dec 23
renFIL	Filecoin	Apr 21	Apr 21–Nov 22			
SNX	Synthetix Network	Dec 20	Dec 20–Nov 22		Dec 20–Jun 23	Dec 23
stETH	Liquid Staked Ether	Feb 22			Feb 22–	
sUSD	Synthetix USD	Dec 20	Dec 20–Apr 24			May 24
TUSD	TrueUSD	Dec 20	Dec 20–Jul 23	Dec 20–Jul 23	Dec 20–Jul 23	
UNI	Uniswap	Nov 20	Nov 20–Nov 22		Nov 20–Jun 23	Dec 23
USDC	USD Coin	Dec 20	Dec 20–	Dec 20–Nov 23	Dec 20–	
USDP	Pax Dollar	Jul 21	Jul 21–Apr 24			May 24
USDT	Tether USD	Nov 20	Nov 20–	Nov 20–Nov 23	Nov 20–Dec 20	
UST**	TerraUSD	Mar 22	Mar 22–May 22			Dec 23
WBTC	Wrapped Bitcoin	Nov 20	Nov 20–	Nov 20–Nov 23	Nov 20–	
WETH	Wrapped Ether	Nov 20	Nov 20–	Nov 20–Nov 23	Nov 20–	
xSUSHI	SushiBar	Feb 21	Feb 21–Nov 21		Feb 21–Nov 22	Dec 23
YFI	yearn.finance	Nov 20	Nov 20–Nov 22	Nov 20–Dec 20	Nov 20–Nov 22	Dec 23
ZRX	0x Protocol	Nov 20	Nov 20–Oct 22	Nov 20–Oct 22	Nov 20–Nov 22	Dec 23

*not / **only partially included in liquidation probability analysis due to availability of Binance spot price data.

For asset market prices the study relies on historical price data from Binance, the largest centralized exchange both in terms of volume and listed assets during the observed period. Specifically, <ASSET>/USDT spot price data at second intervals are used, sourced via the Binance public API. Due to partial availability of price data, the asset scope of the liquidation probability analysis is restricted to 30 out of the 37 reserve assets in Aave V2. Table 1 provides an overview of reserve assets, their roles over time, and inclusion in the liquidation-probability sample.

In addition, block-level gas price information is used as an indicator for current network congestion and cost of liquidation transactions. Gas price data and block timestamps used to align Binance asset prices with block intervals are sourced from the *Google BigQuery* public dataset for Ethereum. For blocks after the Ethereum Merge, the *beaconcha.in* API is used to identify whether its content was sourced via an MEV-boost relay, indicating that transaction selection and ordering in that block was performed by a professional builder maximizing miner-extractable-value (MEV).

Table 2: Summary statistics of independent variables used in the logistic regression analysis of liquidation probability presented in Section 4.2. Basis are positions posing viable liquidation opportunities with a liquidatable volume of at least $\theta = 250$ fin. The dependent variable is liquidation success in the next block, i.e., the position is liquidated at least once.

Variable	Mean	SD	Min	1st Q	Median	3rd Q	Max
Log Liquidation Volume (fin)	2.854	4.683	2.398	2.476	2.576	2.748	8.094
Liquidator Bonus (bp)	545.819	136.361	400.000	500.000	500.000	500.000	1500.000
Adjusted Bonus (bp)	552.941	184.044	-2457.000	463.000	511.000	575.000	3620.000
Oracle Distortion	1.012	0.190	-5.247	0.922	1.002	1.088	6.862
Median Gas Price (gwei)	88.316	133.905	0.000	30.000	64.000	116.000	20421.000
Market Price Move (bp, t-25)	92.756	345.691	0.000	4.000	16.000	41.000	5653.000
30-Day Volatility	0.011	0.005	0.000	0.008	0.010	0.013	0.045
Stablecoin Collateral	0.134	0.341	0.000	0.000	0.000	0.000	1.000
Stable Rate Debt	0.194	0.396	0.000	0.000	0.000	0.000	1.000
MEV Boost (t+1)	0.150	0.357	0.000	0.000	0.000	0.000	1.000

Note: All variables have the same number of observations: 165 225 988.

4 Empirical Analysis and Results

Based on the proposed state model and the dataset collected, this section presents two analyses. Initially, transitions within the introduced state model are summarized including a numerical analysis of transition probabilities. This creates the basis for the subsequent main analysis of the factors associated with liquidation probability, using discrete-time logistic regression with a variety of model specifications.

4.1 Transition Probabilities

The transition probability analysis examines how borrower positions transition between states. The resulting patterns provide descriptive evidence on the frequency, direction, and drivers of state changes.

Let P denote the one-block transition probability matrix for borrower positions active in Aave V2 during the sample period, using the baseline threshold $\theta = 250$ fin:

$$P = \begin{bmatrix} P_{HH} & P_{HV} & P_{HS} \\ P_{VH} & P_{VV} & P_{VS} \\ P_{SH} & P_{SV} & P_{SS} \end{bmatrix} = \begin{bmatrix} > 0.99 & 3.08 \cdot 10^{-7} & 2.11 \cdot 10^{-6} \\ 2.02 \cdot 10^{-4} & > 0.99 & 2.81 \cdot 10^{-5} \\ 1.16 \cdot 10^{-5} & 1.60 \cdot 10^{-7} & > 0.99 \end{bmatrix}$$

Each matrix entry gives the probability for a position in a given state at block t to transition to the next state at $t + 1$. For example, P_{HV} and P_{HS} capture the probabilities that in the next block a healthy position transitions into (V) or (S) respectively. Thereby, P_{HH} represents the probability for the position to remain in (H). By construction, transition probabilities originating from the same state sum to 1.

Transitions between states are rare at block frequency. When they occur, (H) most often is either origin or destination, while direct transitions between the two unhealthy states (V) and (S) are comparatively infrequent. When exiting (H), a substantial proportion transitions directly into (S). At the same time, the majority of positions leaves (S) through restoring health. Compared to (V) as starting point, positions in (S) are less likely to be restored to health, consistent with the interpretation of (S) capturing smaller size positions unattractive for liquidation.

For a more detailed look at positions in (V), the data distinguishes between the three transition causes described in Figure 5. If a position is liquidated or adjusted by the borrower in the next block, the transition is attributed accordingly. In the very rare case that both occur for the same position, the liquidation is given precedence. This reflects that a borrower action rationally should restore position health and thereby prevent a later liquidation in the

same block. If neither the borrower nor liquidators become active, the position's transition is attributed to changes in market conditions. This decomposition splits the second row of the initial transition probability matrix into additive components, $\sum_q P_{Vq} = \sum_q (P_{Vq}^m + P_{Vq}^b + P_{Vq}^l)$ with $q \in \{H, V, S\}$, yielding:

$$P' = \begin{bmatrix} P_{HH} & P_{HV} & P_{HS} \\ P_{VH}^m + P_{VH}^b + P_{VH}^l & P_{VV}^m + P_{VV}^b + P_{VV}^l & P_{VS}^m + P_{VS}^b + P_{VS}^l \\ P_{SH} & P_{SV} & P_{SS} \end{bmatrix}$$

$$= \begin{bmatrix} > 0.99 & 3.08 \cdot 10^{-7} & 2.11 \cdot 10^{-6} \\ (14.12 + 0.21 + 5.90) \cdot 10^{-5} & > 0.99 + (0.28 + 16.51) \cdot 10^{-6} & (2205.66 + 0.56 + 607.22) \cdot 10^{-8} \\ 1.16 \cdot 10^{-5} & 1.60 \cdot 10^{-7} & > 0.99 \end{bmatrix}$$

Notably, borrower interventions on positions in (V) are rare. When they occur, they almost exclusively restore the position's health. Changes in market conditions account for approximately two-thirds of exits from (V), with liquidations responsible for the remainder. The data for P_{Vx}^l allow for two main observations. First, about 72% of positions in (V) that are liquidated end up healthy by the end of the block. Second, among the remaining liquidation-caused transitions, P_{VV}^l considerably exceeds P_{VS}^l . This pattern is consistent with partial-liquidation sequences in Aave V2 terminating before the remaining liquidatable amount falls below θ . Further, the data shows it to be materially more likely for a VS transition to be caused by market changes than by liquidators. At the same time, the baseline θ already classifies 26.2% of the 32 658⁵ liquidation cases as having occurred on already stale positions. With the potential concern of result sensitivity to θ in mind, the robustness of main analysis results to an alternative threshold is examined in Section 5.

4.2 Liquidation Probability

In the main analysis, the focus is on factors associated with the probability that a viable liquidation opportunity is liquidated in the next block $t+1$. To this end, the following discrete-time logistic regression model with monthly time-fixed effects is estimated:

⁵Out of the 33 168 liquidations in Aave V2 over the period of investigation, 510 liquidations occurred on positions that have already been liquidated before in the same block.

$$\begin{aligned}
\ln \left(\frac{P_{HH}^l + \sum_q P_{Vq}^l}{\sum_q (P_{Vq}^m + P_{Vq}^b)} \right)_{t+1} &= \beta_0 + \beta_1 \text{ Log Liquidation Volume}_t \\
&+ \beta_2 \text{ Liquidator Bonus}_t + \beta_3 \text{ Adjusted Bonus}_t \\
&+ \beta_4 \text{ Oracle Distortion}_t + \beta_5 \text{ Median Gas Price}_t \\
&+ \beta_6 \text{ Market Price Move (t-25)}_t \\
&+ \beta_7 \text{ 30-Day Volatility}_t + \beta_8 \text{ Stablecoin Collateral}_t \\
&+ \beta_9 \text{ Stable Rate Debt}_t + \beta_{10} \text{ MEV Boost}_{t+1} + \gamma_m
\end{aligned} \tag{5}$$

The analysis expands beyond positions in (V) to include P_{HH}^l , i.e., liquidations that occurred in the same block the underlying position became unhealthy and thus were not (yet) in (V) at t . This choice increases the number of observed liquidation cases from 13 306 to 22 906 and improves estimation precision. The reference comprises all positions in (V) at t , but not liquidated in $t + 1$, regardless of whether they transition out or remain in this state.

Liquidations occur with extreme relative rarity in the over 165 million high-frequency observations. Due to this severe class imbalance and for computational tractability, a stratified control undersampling approach is followed. Given the liquidation cases in an observation month, non-liquidated controls are randomly sampled at a fixed case-control ratio of 1:20. An analysis of coefficient stability for different ratios is shared in Appendix C. While presented baseline probabilities are adjusted using the intercept correction by King and Zeng (2001) and undersampling controls may have no effect on parameter estimation efficiency (Wang, 2020), the discussion of results focuses primarily on the direction and statistical significance of estimated coefficients.

The results for different model specifications within Equation 5 are summarized in Table 3, showing regression coefficients with standard errors in brackets. Note that for numerical stability in estimating the models, all non-dummy variables with the exception of **Log Liquidation Volume** are rescaled by their interquartile range. Further, to account for potential within position dependence of observations, cluster-robust standard errors are reported with the position as clustering unit.

The six model specifications are estimated with a progressively richer set of variables. Models (1) through (4) explore the effects of liquidation incentives and transaction cost. Thereby, special focus is given to whether and in what form potential distortions between oracle and market asset prices have an effect. Model (5) introduces variables linked to asset price volatility to shed light on the potential impact of price risk on liquidation decisions. Finally, model (6) adds a dummy to distinguish between variable and stable interest rate positions. All specifications control in the form of **MEV Boost (t+1)** for the impact of a block being sourced from

Table 3: Logistic regression estimates of next-block liquidation probability for viable liquidation opportunities in the Aave V2 Main Market on Ethereum. This table reports coefficient estimates for the probability of liquidation in the next block $t+1$ for borrower positions that satisfy $\bar{v}^{(k)} \geq \theta$ at block t . The dependent variable equals one if the position is liquidated at least once in the subsequent block and zero otherwise. The estimation sample includes all liquidation cases where $\bar{v}^{(k)} \geq 250$ fin at t . Controls are undersampled from the population of all borrower positions in state (V) at t that are not liquidated during $t+1$, maintaining a 1:20 case-control ratio within monthly bins. All specifications include monthly time-fixed effects and intercepts are corrected for control undersampling following King and Zeng (2001). Cluster-robust standard errors are reported in parentheses.

Logistic Regression with Time-fixed Effects for Probability of Liquidation in $t+1$ Reference = No Liquidation. 481 026 observations with $\bar{v}^{(k)} \geq 250$ fin covering 22 906 cases with controls randomly sampled out of 165 203 082 for a 1:20 case-control ratio in each month (46 bins).						
	(1)	(2)	(3)	(4)	(5)	(6)
Log Liquidation Volume	4.28425*** (0.07327)	4.23678*** (0.07227)	4.24533*** (0.07321)	5.32703*** (0.11311)	5.57755*** (0.08202)	5.57587*** (0.08207)
Liquidator Bonus	0.00276*** (0.00017)		0.00283*** (0.00017)	0.00332*** (0.0002)	0.00356*** (0.0002)	0.00356*** (0.0002)
Adjusted Bonus		0.06606*** (0.01165)				
Oracle Distortion			-0.10156*** (0.01352)	-0.08031*** (0.01517)	-0.09152*** (0.01392)	-0.09159*** (0.01392)
Median Gas Price				-0.5074*** (0.02524)	-0.59711*** (0.00413)	-0.59694*** (0.00421)
Price Move (t-25)					0.01171*** (0.00339)	0.01172*** (0.0034)
30-Day Volatility					-0.56501*** (0.05497)	-0.56422*** (0.05507)
Stablecoin Collateral					-2.58932*** (0.61139)	-2.58929*** (0.61147)
Stable Rate Debt						-0.03138 (0.08587)
MEV Boost (t+1)	1.61318*** (0.16296)	1.56712*** (0.16414)	1.63143*** (0.16341)	1.77489*** (0.19249)	1.80936*** (0.16552)	1.80886*** (0.16553)
Intercept	-25.59198*** (0.57568)	-23.66783*** (0.50928)	-24.89696*** (0.58537)	-28.7511*** (0.74589)	-28.04002*** (1.09888)	-28.02894*** (1.09949)
$R^2_{McFadden}$	0.277	0.268	0.28	0.312	0.338	0.338
R^2_{Tjur}	0.077	0.072	0.079	0.111	0.128	0.128

*p<0.1; **p<0.05; ***p<0.01

professional builder networks on liquidation probability.

Consistent with intuition, liquidation success is strongly associated with economic incentives. Larger volumes and higher liquidation bonuses are both strongly and significantly associated with increased probability of liquidation success throughout model specifications. In contrast, blocks with increased gas prices are associated with lower liquidation success in the next block. This is consistent with liquidators responding rationally to arbitrage opportunities net of transaction cost.

With regard to the role of potential oracle deviations, results across models (1)–(3) suggest that **Liquidator Bonus** as parametrized in the lending market is a superior predictor of liquidation success than the alternative **Adjusted Bonus**, which corrects the bonus parameter for current deviations between oracle and market asset prices. At the same time, accounting for oracle deviations in the form of the separate **Oracle Distortion** factor, further improves the adjusted R^2 s. Its negative coefficient, however, associates oracle deviations in favor of liquidators with lower likelihood of liquidation in the next block. This pattern is consistent with liquidators accounting for oracle deviations, with competing sub-effects involved. On one hand, distortion at t affects relative profitability outlook for a liquidation at $t + 1$. On the other hand, increased deviations are associated with update pressure for oracles (see, e.g., Nadler et al., 2026), which implies increased execution risk for a liquidation in $t + 1$. Since, conceptually, oracle updates bring **Oracle Distortion** towards 1, values exceeding 1 at t reflect increased risk of negative shifts in relative profitability for $t + 1$ and the reverse case whenever **Oracle Distortion** is below 1 at t .

Model (5) shows price volatility variables are significantly correlated with likelihood of liquidation in the next block. Increased latent volatility between collateral and debt asset prices is associated with lower probability of liquidation. This pattern is consistent with higher price volatility implying increased execution risk over short horizons. Notably, the association with short term price moves is positive. This aligns with the hypothesis that shifts to expected profitability caused by short-term price moves potentially offset increased execution risk. Further notable is the correlation of liquidation success with non-stablecoin collateral. If liquidators' unit of account are in a fiat currency, a preference for liquidating positions with stablecoin collateral could be a viable hypothesis, as it potentially reduces the exchange steps required to complete the arbitrage cycle and therefore transaction cost and price risk. The significant negative effect of the **Stablecoin Collateral** dummy, however, rejects this hypothesis.

The interest rate model underlying a position has no effect on liquidation process or economic incentives in Aave V2. However, the mechanics of monitoring variable and stable rate debt positions differ slightly. Assuming certain fixed cost for set-up and maintenance of monitoring infrastructure, this may give rise to heterogeneous groups of potential liquidators. However, the lacking significance of the result for the **Stable Rate Debt** dummy fails to associate the

interest model with liquidation probability.

At the same time, liquidations are significantly more likely to occur in blocks built through MEV-boost relays. This pattern is consistent with the hypothesis that proposer-builder-separation (PBS), where the block’s proposer auctions off the privilege to select and order its transactions to a network of professional builders, may play a role in the likelihood of liquidation success. PBS adds new possibilities for liquidators to reduce execution risk, such as private transaction flows or transaction bundling (see, e.g., the study on PBS by Heimbach et al., 2023). However, it must be acknowledged that PBS has a multitude of effects on transaction inclusion. In addition, **MEV Boost ($t+1$)** may also capture in part shifts in liquidation probability associated with the Ethereum Merge in September 2022. This change in consensus model of the Ethereum network was a prerequisite for PBS to be established and around 8 out of 9 post-Merge blocks during the analysis period have been built through MEV-boost relays.

The estimated monthly fixed effects with baseline December 2020 for the six model specifications are presented in Figure 7. While the associated effects appear generally stronger for more richly specified models, the major trends appear the same. Notably, a marked shift in baseline liquidation probabilities is observed in the fourth quarter of 2022. The initial phase up to October 2022, coinciding with Aave V2’s peak activity in both debt volume and variety of reserve assets, exhibits significant positive log-odds shifts across models. This phase ends abruptly in November 2022, when shifts first become negative, then return to a persistently lower level that is hardly ever significant.

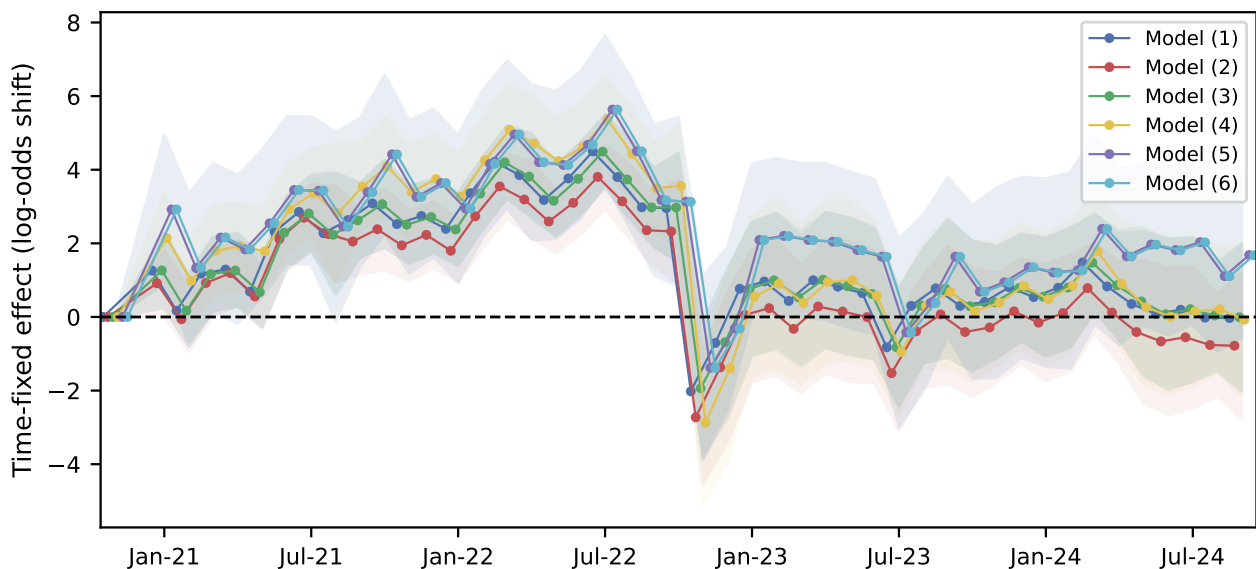


Figure 7: Estimated monthly time-fixed effects with 95% confidence intervals for each model specification in main analysis of liquidation drivers. December 2020 serves as baseline month. Note: Lines for different model specifications are slightly horizontally offset to improve readability. This offset has no analytical meaning.

While the macro factors potentially influencing liquidation probability are manifold, several events coincide with this shift. First, the consensus mechanism shift by Ethereum with the Merge in September 2022, albeit some of that impact conceptually should be captured by **MEV Boost** ($t+1$). Second, the launch of Aave V3 on Ethereum in January 2023 in combination with the decommissioning of several reserve assets for lending in Aave V2 during October and November 2022, as shown in Table 1.

In summary, while baseline probabilities may shift over time, the results indicate that liquidation success at block frequency is primarily associated with economic incentives and transaction costs, with additional frictions associated with oracle distortions, price volatility, and block sourcing. These findings motivate the robustness analyses presented in the next section.

5 Robustness

This section examines the robustness of the empirical results presented in Section 4.2. It assesses the stability of the main findings under an alternative definition of viable liquidation opportunities, as well as for subsamples that split the data by interest model type of the position or the Ethereum consensus regime at the time. In addition, it remarks on potential sources of endogeneity and the employed mitigation strategies. The robustness checks are intended to assess the stability of the estimated associations rather than to establish causal effects.

5.1 Subsamples

As acknowledged, the classification of viable liquidations requires choosing θ . The first subsample analysis hence examines the consistency of the logistic regression results under a stricter threshold. Table 4 and Figure 11 in the appendix report the logistic regression estimates using $\theta = 500$ fin, reducing the total population of observations to 49 841 339 including 19 231 liquidation cases. The results indicate that the effect directions and significance levels of the main analysis largely hold also under a stricter definition of viable liquidation opportunities. Two exceptions are **Adjusted Bonus** and **Price Move** ($t-25$) that exhibit lower significance, consistent with reduced sensitivity to relative profitability when smaller liquidation opportunities are excluded. Overall, the results indicate that the main findings are not driven by marginal liquidation events near the baseline cutoff.

Second, the data are split into stable rate and variable rate debt positions. This explores whether the baseline results hold equally. The regression outputs for each subset are shown in Tables 5 and 6 in the appendix. Notably, stable-rate liquidations account for approximately one-sixth of all liquidation cases and are observed over a shorter time horizon due to the gradual phase-out of stable borrowing in Aave V2. While disablement of stable borrowing prevents new positions being opened, existing ones can be maintained. The data contains the last stable rate

position liquidation in May, 2024, curtailing the observation period accordingly. While variable rate subset results are in line with the baseline results, the stable rate subset shows some differences. Particularly, the volatility measures added in model (5) are no longer significant, with **30-Day Volatility** and **Stablecoin Collateral** even switching sign. In addition, the results show a decrease in significance level for the **MEV Boost (t+1)** dummy variable under model specifications (1) and (2). Relevant context here is that the dummy is much rarer in the stable rate subset, i.e., taking the value of 1 in only 4.0% of observations considered compared to 12.7% in the baseline context. As significance levels rise again with more richly specified models and given the considerably smaller number of observations, the lowered significance may be due to conditions causing less precise estimates.

Lastly, the dataset is split into observations before and after the Ethereum Merge on September 15, 2022, to test whether the consensus mechanism change is associated with differences in liquidation dynamics. Although the Merge cuts the observation period almost perfectly in half, more than 85% of the liquidation cases fall into the pre-Merge period. The regression outputs are shared in Tables 7 and 8 with the **MEV Boost (t+1)** dummy excluded from the variable set, as PBS is only feasible under Ethereum’s new consensus mechanism. Naturally, there are fewer monthly bins in each estimate, with an overlap only for September 2022. Most of the effect direction and significance results of the baseline analysis prevail in either subsample. One exception is the reduced significance of **Price Move (t-25)** in both subsets. The **Median Gas Price** coefficient notably loses significance and switches direction under model specification (4) in the post-Merge subsample. While it switches back and regains significance in the more richly specified models and the subsample is again relatively small, a weakened signal from **Median Gas Price** would be consistent with the new consensus model opening parallel avenues for remunerating block builders for transaction inclusion.

In summary, the subsample analyses reveal no significant change in effect direction and to a large extent support the findings of the main analysis. Some coefficient estimates have lower significance than in the baseline. This occurs more often in the smaller subsets, where estimation power is inherently lower. The repeatedly lower significance of **Price Move (t-25)** may indicate a less robust association with liquidation success across subsamples.

5.2 Endogeneity

Acknowledging possible sources of endogeneity, this subsection lays out the empirical strategies employed to mitigate bias from reverse causality, simultaneity, measurement error or omitted variables. It further describes measures taken to support robust statistical inference.

In terms of reverse causality, the analysis setup inherently gives limited cause for concerns. Primarily due to the temporal structure, where all but one explanatory variable are predetermined at the time of block t relative to the outcome in the next block $t + 1$. **MEV Boost (t+1)**

constitutes the only exception from this lagged setup. However, whether the validator at $t + 1$ outsources block-building is unlikely to be driven by individual liquidation opportunities.

With regard to simultaneity, it can be reasonably assumed that liquidators generally do not influence position characteristics at t . However, it must be acknowledged that Lehar and Parlour (2023) find anecdotal evidence of predatory liquidations, where liquidator addresses short sell a collateral asset shortly before a liquidation wave ensues. Such behavior is not expected to be systematic and therefore unlikely to materially affect the aggregate associations estimated in this study.

High-frequency data typically warrant heightened concern with regard to measurement errors. Accuracy of variables derived from on-chain data sourced directly from Ethereum archive nodes is generally ensured. Other network related data are sourced from reputable sources, such as Google BigQuery and beaconcha.in. For market price information, the reliance on the public Binance API and the potential inaccuracies from relying on a single price source capturing only a fraction of the market and centralized crypto exchanges which may be affected by wash trading (see, e.g., Cong et al., 2023) must be acknowledged. Nevertheless, Binance represents the most liquid and comprehensive spot market during the sample period by a considerable margin, making it a pragmatic benchmark.

Potential omitted-variable bias is mitigated through a rich set of controls capturing liquidation incentives, transaction costs, price dynamics, and network conditions. Moreover, monthly time-fixed effects absorb common shocks and protocol-level changes over time. Given the pseudonymous nature of blockchain addresses, user-specific characteristics are largely unobservable and therefore unlikely to systematically confound position-level liquidation outcomes. The stability of coefficient signs and significance across progressively richer specifications further supports this interpretation.

Finally, as positions may go through multiple unhealthy episodes over time, the data contains multiple observations from the same positions. Unobserved shared characteristics of these observations are accounted for via cluster robust standard errors with the position address as clustering identifier. Thus, potential overstatement of statistical significance is mitigated in light of possible within-position dependence and heteroskedasticity.

6 Conclusion

Collateral constitutes the central mechanism of credit risk management in decentralized finance. In the pseudonymous DeFi environment, the loss of collateral is the primary consequence borrowers face upon default. Instead of reputation-based lending and credit risk scores, DeFi lending relies on overcollateralization and efficient liquidations whenever protocol risk constraints are violated. Frictions in the liquidation process may lead to accumulation of bad debt in DeFi

protocols, threatening their economic viability and eventually the equity of liquidity providers. Recent events in DeFi lending illustrate that, once bad debt materializes, loss absorption may become a concrete governance and balance sheet problem rather than a pure technical concern.

This paper examines liquidation dynamics in DeFi lending markets, a highly successful category of blockchain-based financial applications. Focusing on the 54127 borrowing positions between November 2020 and September 2024 in the Aave V2 Main Market on Ethereum, it examines how undercollateralized positions evolve and what factors are associated with unsuccessful or delayed liquidations in one of the largest and most mature lending protocols. Proposing a Markov-like state model, it classifies positions as healthy, viable for liquidation, or stale, based on their collateralization relative to protocol risk parameters and size of their inherent liquidation opportunity.

Using a novel dataset that tracks borrower position states at block frequency, i.e., at approximately 12 second intervals, two empirical questions are explored. First, the analysis documents state transition probabilities of borrower positions, with particular attention to viable liquidation positions that are not exploited immediately. Second, it analyzes which position characteristics, market conditions, and network features are associated with the probability that a viable position is liquidated in the subsequent block.

Descriptive evidence indicates that liquidations in Aave V2 are generally timely when assessed at an aggregate level. 95.9% of insufficiently collateralized debt is liquidated within the first minute of risk parameter violations. Also, volume of debt open for liquidation never exceeds 0.25% of total outstanding debt at the end of each day. However, albeit at low levels, open liquidations appear to persist in the market and 23.5% of liquidation events occur on positions that have been open for more than one day. The transition probability analysis further shows that once a position enters an insufficiently collateralized state, borrowers rarely intervene. The results further document that, for positions that are not liquidated immediately but are classified as viable liquidation opportunities, the probability of health being restored through changes in lending market conditions, such as oracle price updates, is about 1.7-times larger than the position undergoing liquidation. These patterns highlight heterogeneity in liquidation dynamics warranting further exploration.

Logistic regression evidence on next-block liquidation probability highlights the role of incentives, costs, and execution risk. Liquidation success is strongly associated with size and relative profitability of the inherent arbitrage opportunity. Consistent with the notion of economic incentives playing a key role, transaction costs in the form of network gas prices are associated with a lower likelihood of liquidation. Results further reveal patterns consistent with deviations between oracle and market asset prices as well as price volatility between collateral and debt asset as source of liquidation frictions. In addition, liquidations appear to be more likely to occur in blocks with proposer-builder-separation through MEV-boost relays,

albeit this association must be interpreted with caution in light of potential confounding factors.

Overall, the findings contribute empirical evidence from high-frequency data on the microstructure of DeFi liquidations, a central component of credit risk management in decentralized lending markets. While the analysis is limited to a single protocol, the results highlight mechanisms through which liquidation frictions may arise even in mature DeFi systems. These insights may inform future research on DeFi market design, liquidation incentives, and risk management, as well as ongoing discussions among protocol designers, practitioners, and policymakers.

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A Liquidations in DeFi

Figure 8 illustrates fundamental economics of liquidations for borrowers and protocols in DeFi, assuming no network or protocol fees or liquidator bonuses. $D^{(k)}$ represents the current value of the assets deposited with the protocol on account of blockchain address k . Complementarily, $B^{(k)}$ denotes the current value of the assets that have been borrowed. Rationally, borrowers will honor their liabilities as long as $D^{(k)} > B^{(k)}$, i.e., as long as their position equity and therefore their payoff from closing the position is positive. In case asset price changes or accruing interest cause the current value of $D^{(k)}$ to fall below $B^{(k)}$, there are no consequences for the borrower to abandon position k , leaving the protocol with bad debt in the amount of $B^{(k)} - D^{(k)}$. The permissionlessness of DeFi protocols implies that there is no identification of the entity behind k as prerequisite for protocol use, i.e., recourse on any borrower assets, that are not locked with the protocol, is effectively not possible. Further, if the rationally defaulting borrower behind k wants to continue using the same protocol, they can simply do so using a different blockchain address $\neq k$.

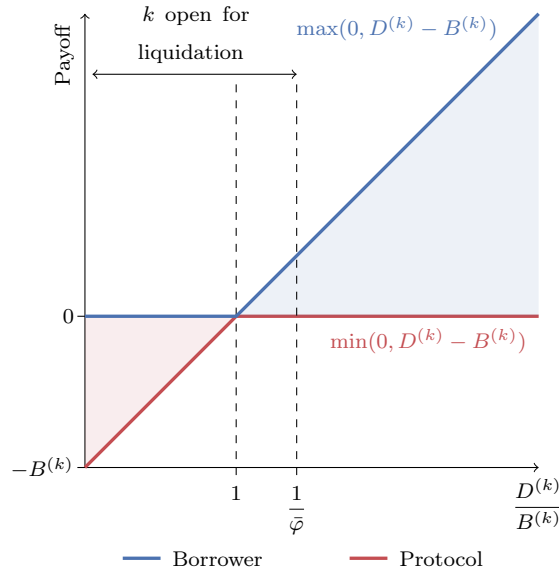


Figure 8: Illustrating the payoff of closing a collateralized debt position k from the borrower and protocol perspective, depending on the value of the positions deposited collateral, $D^{(k)}$ relative to the amount borrowed, $B^{(k)}$, assuming no transaction cost or protocol fees. k is open for liquidation, whenever collateralization $\frac{D^{(k)}}{B^{(k)}}$ falls below the threshold defined by the maximum loan-to-value ratio specified in the protocol $\bar{\varphi}$. Illustration assumes no transaction and protocol fees or liquidator bonuses in case of liquidation.

To effectively manage credit risk whenever the price of collateral assets decline, protocols with lending components must force the closing of any position k , before a state of $D^{(k)} < B^{(k)}$ is reached. For this purpose, DeFi protocols generally require overcollateralization, i.e., they maintain maximum loan-to-value parameters $\bar{\varphi} < 1$. Whenever k no longer satisfies the required

collateralization, it is open for liquidation. In this case, the assets underlying $D^{(k)}$ are for sale with the proceeds going towards repayment of $B^{(k)}$. For context, recall that as mere smart contracts, protocols cannot actively close a position themselves. Further, the pseudonymous nature of blockchain addresses precludes any notification of the borrower behind k in a margin-call-like fashion. Instead, protocols rely on third party liquidators to become active and close insufficiently collateralized positions for them. Generally, liquidations are open to anyone and, while designs can vary significantly, incentives rely on the sale of collateral at a discounted price, making open liquidations arbitrage opportunities.

B Variable Description

This section provides the explanation and describes the construction of the variables employed in the main analysis in Section 4.2 as well as the corresponding robustness analyses discussed in Section 5. The variables combine a selection of observable position attributes with established variables for evaluating market conditions of the debt and collateral assets that are open for liquidation. Thereby it is assumed that the largest debt and collateral asset of a position are considered for liquidation. Variables are measured at the position-block pair level and, with the exception of **MEV Boost (t+1)**, used with a lag of one block. Summary statistics are shared in Table 2.

Log Liquidation Volume (fin) The maximum liquidation volume in terms of repaid debt for the unhealthy position denominated in finney or 10^{-3} Ether. It is constructed based on the position’s debt and collateral asset balances, valued at the Aave V2 oracle price, and accounts for the constraints described in Equation 4. For the purpose of this study, it is used both as measure to classify which positions open for liquidation are viable and as measure for the size of the arbitrage opportunity in case of a liquidation in the following block.

Liquidator Bonus (bp) The liquidation bonus ϕ_j of the position’s largest collateral asset j as per the Aave V2 market parameters in the form of percentage of debt repaid. Represents the profit before any transaction cost and ignoring potential deviations between the oracle and market prices for the assets involved in the liquidation.

Adjusted Bonus (bp) The liquidation bonus adjusted for current deviations between Aave V2 oracle and market asset prices as percentage of debt repaid. Formally defined as $\dot{\phi}_{i,j} = \left(\frac{\dot{p}_j \cdot (1 + \phi_j)}{p_j} - \frac{\dot{p}_i}{p_i} \right)$ with the collateral j and debt i asset defined by the largest holdings of the position and $\dot{p}$ and p the corresponding Binance market and Aave V2 oracle prices. Represents the profit before any transaction cost accounting for deviations between the oracle and market prices for the assets involved in the liquidation.

Oracle Distortion Factor Factor of the distortion of potential oracle price deviations as the multiplier between **Liquidator Bonus** and **Adjusted Bonus**, measured as $\frac{\dot{\phi}_{i,j}}{\dot{\phi}_j}$. While a factor of 1 reflects absence of oracle price distortions, values exceeding 1 represent times where deviations between market and oracle prices of the collateral and debt asset are in favor of the liquidator, and the opposite when the factor is below 1.

Median Gas Price (gwei) Median price per billion units of gas paid in this block in gwei, i.e., 10^{-9} Ether, disregarding zero fee transactions. Total gas prices are considered, i.e., base plus priority fees after the Merge. Gas prices are both a direct, liquidation volume independent transaction cost for liquidators and a reliable indicator of current block space demand and network congestion.

Price Move (bp, t-25) Short time market price changes in basis points of the collateral in terms of the debt asset, with the assets determined by the largest holdings of the position in each category. Formally defined as $|\frac{p_{j,t} \cdot \dot{p}_{i,t-25}}{p_{j,t-25} \cdot \dot{p}_{i,t}} - 1| \cdot 10^4$, where $\dot{p}$ represent the market prices taken from the Binance exchange. With the lag of 25 blocks corresponding to approximately 5 minutes, this measure serves as a short-term price volatility indicator.

30-Day Volatility The mid-term hourly price volatility between the main collateral and debt asset of the position. Based on Binance price data using log returns over a rolling 30-day window ending with the last full hour before the block. This measure serves as mid-term price volatility indicator.

Stablecoin Collateral Dummy variable equal to 1 if the position’s main collateral asset can be classified as a fiat-pegged stablecoin. This measure serves as indicator whether liquidators receiving stablecoin collateral systematically impacts the probability of liquidation.

Stable Rate Debt Dummy variable equal to 1 if the position’s main debt is borrowed under Aave V2’s stable interest rate model. Interest accrual mechanics for variable and stable debt differ and in hence require different monitoring setups from liquidators.

MEV Boost (t+1) Dummy variable equal to 1 if the following block has been sourced from a MEV-boost relay. After the Ethereum Merge, MEV-boost relays enable Proposer-Builder-Separation (PBS). They allow validators to outsource the building of a block to a network of professional builders, who compose a block maximizing their own miner extractable value and bid through the relay for inclusion in the next slot. This control variable accounts for the potential impact of block building marketplaces on liquidation probabilities in post-Merge Ethereum.

C Result Stability with Undersampling Controls

In light of extreme relative rarity of liquidation events at block frequency, all logistic regression results in this paper are estimated using stratified control downsampling. In each observation month, all liquidation cases are retained, while non-liquidated observations are randomly sampled at a fixed case-control ratio of 1:20, ensuring stability of monthly fixed effects.

To assess the sensitivity of the estimated coefficients to the downsampling approach, this appendix section explores alternative case-control ratios and random sampling variations. For this purpose, the downsampling procedure was repeated 40 times with distinct seeds for six case-control ratios each: 1:5, 1:10, 1:15, 1:20, 1:25, and 1:30. For each of the 240 samples, the six model specifications from the main analysis are re-estimated.

Figure 9 reports mean coefficients and standard errors across the 40 samples with increasing case-control ratios for each variable under the different model specification. Note that all values are scaled with the mean coefficient under the ratio 1:5 to support relative comparison. Across variables, coefficient signs remain stable, with no systematic reversals. Mean coefficients exhibit gradual drift, with changes diminishing at higher ratios. While standard errors often decline as the control sample expands, precision is non-monotone for some variables in richer specifications. Consistent with finite-sample variation and collinearity, these changes do not affect coefficient signs or the qualitative conclusions. Overall, qualitative inference appears stable, although estimation precision does not improve uniformly with increasing ratios.

The distribution of coefficient estimates across each group of 40 samples are summarized in Figure 10. In line with Figure 9, the boxplots show no material skewness or instability and the distribution remains stable or decreases with increasing ratios. This suggests low sensitivity of main results to individual sample draws or ratio choice.

In summary, the results indicate that the qualitative inference is reasonably stable across a wide range of case-control ratios and random control draws. While increasing the ratio generally improves precision, these improvements diminish at higher ratios and may be offset by non-monotone precision for richer model specifications. Consequently, the baseline case-control ratio of 1:20 represents a pragmatic compromise between estimation precision, numerical stability, and computational tractability.

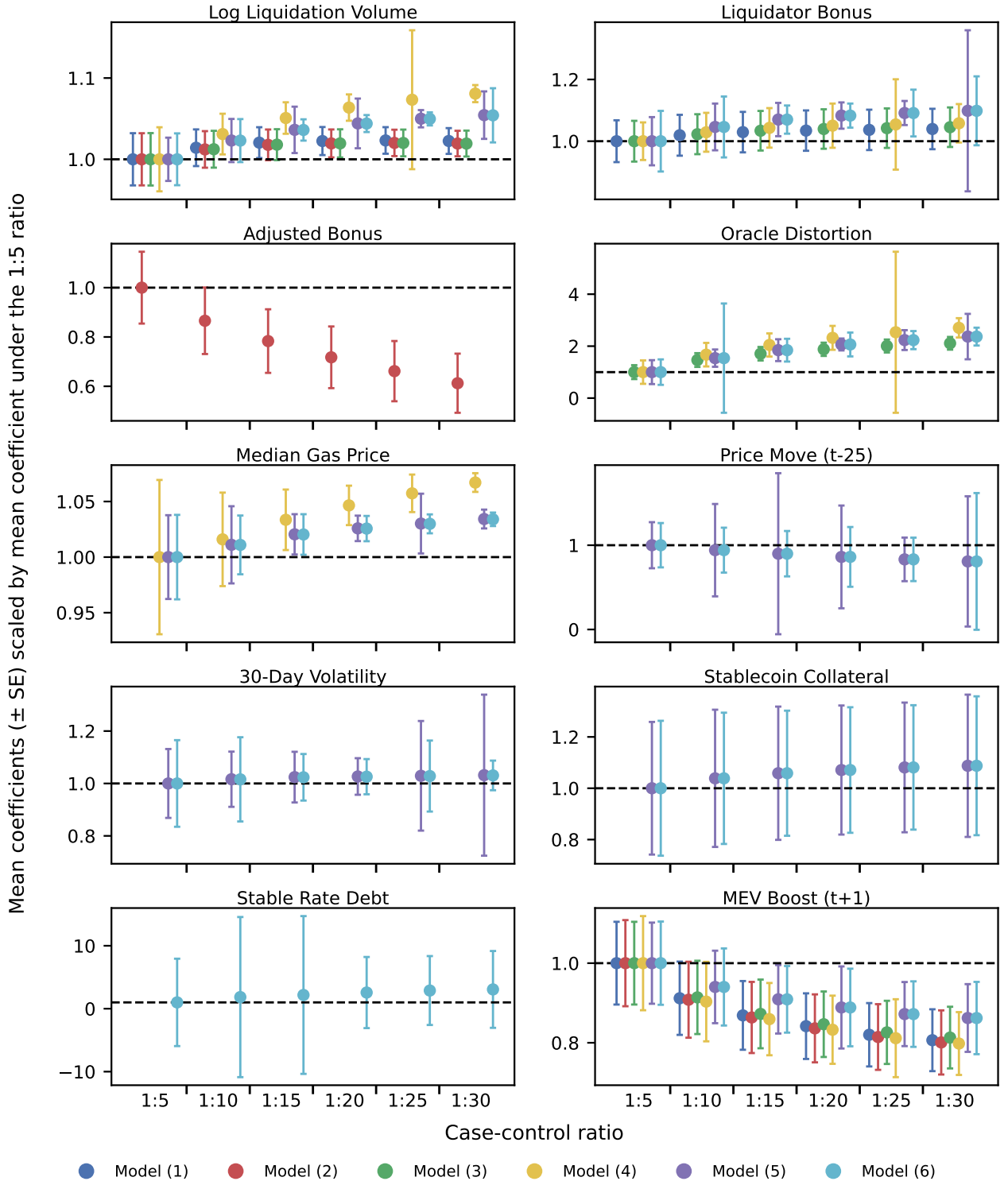


Figure 9: Mean coefficients and SE bands per explanatory variable and model specification with increasing case-control ratios, scaled by the mean coefficient under the 1:5 ratio. Each value is based on results from logistic regressions with monthly time-fixed effects on 40 samples with distinct seeds. Note that variables only show models where they are part of the specification.

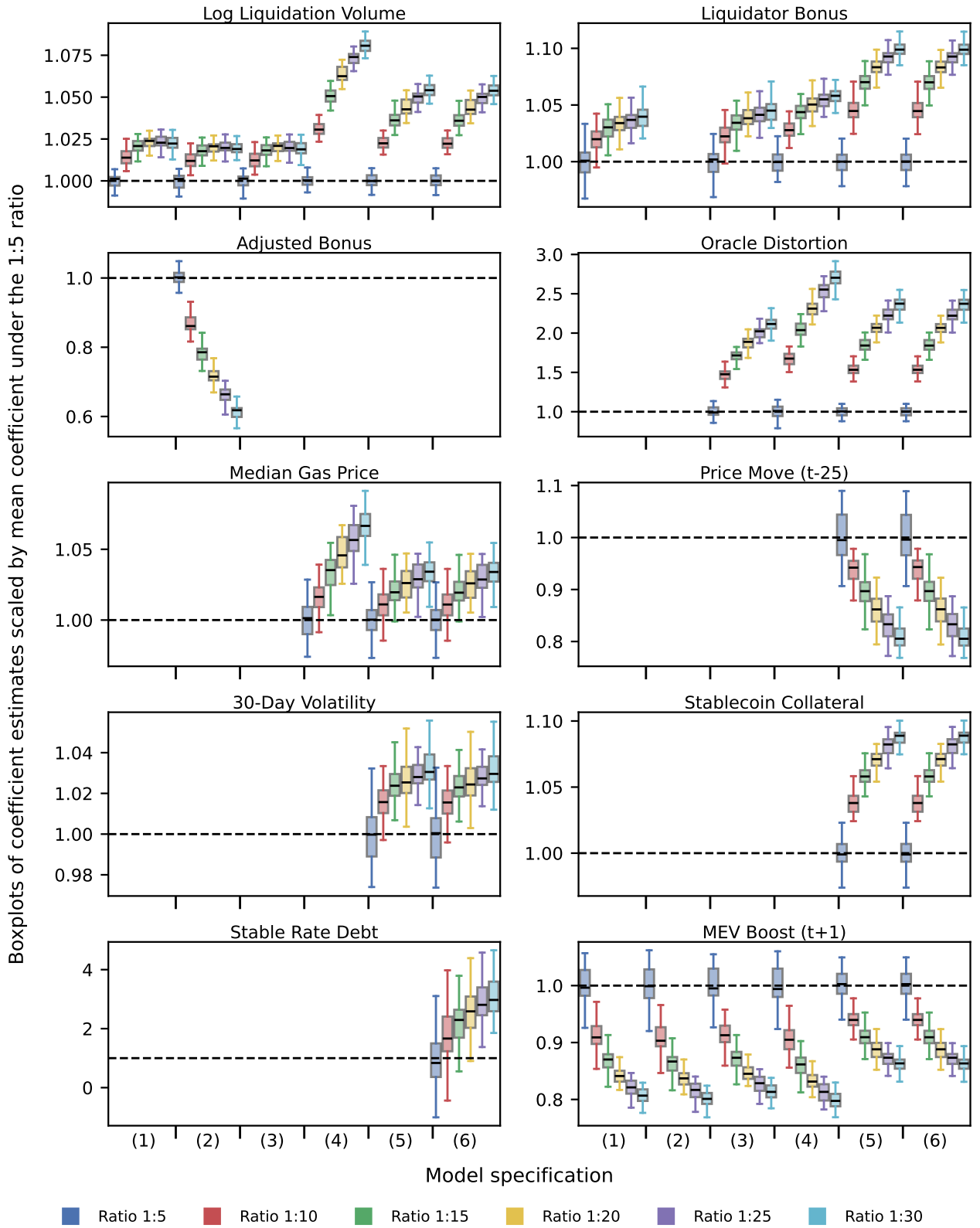


Figure 10: Boxplots of 40 estimated coefficients per variable and ratio with increasingly richer model specification, scaled by the mean coefficient under the 1:5 ratio. Values are based on results from logistic regressions with monthly time-fixed effects on 40 samples with distinct seeds. Note that variables only show models where they are part of the specification.

D Robustness Subsamples

D.1 Increased Cut-off

Table 4: Logistic regression estimates of next-block liquidation probability for viable liquidation opportunities in the Aave V2 Main Market on Ethereum with an increased cut-off. This table reports coefficient estimates for the probability of liquidation in the next block $t + 1$ for borrower positions that satisfy $\bar{v}^{(k)} \geq \theta$ at block t . The dependent variable equals one if the position is liquidated at least once in the subsequent block and zero otherwise. The estimation sample includes all liquidation cases where $\bar{v}^{(k)} \geq 500$ fin at t . Controls are undersampled from the population of all borrower positions in state (V) at t that are not liquidated during $t + 1$, maintaining a 1:20 case-control ratio within monthly bins. All specifications include monthly time-fixed effects and intercepts are corrected for control undersampling following King and Zeng (2001). Cluster-robust standard errors are reported in parentheses.

Logistic Regression with Time-fixed Effects for Probability of Liquidation in $t + 1$ Reference = No Liquidation. 403 851 observations with $\bar{v}^{(k)} \geq 500$ fin covering 19 231 cases with controls randomly sampled out of 49 841 339 for a 1:20 case-control ratio in each month (46 bins).						
	(1)	(2)	(3)	(4)	(5)	(6)
Log Liquidation Volume	4.18549*** (0.08299)	4.1584*** (0.08002)	4.12126*** (0.08334)	5.55666*** (0.12973)	5.63125*** (0.12452)	5.63127*** (0.12406)
Liquidator Bonus	0.00282*** (0.0002)		0.00288*** (0.0002)	0.00339*** (0.00026)	0.00373*** (0.00025)	0.00373*** (0.00025)
Adjusted Bonus		0.02769** (0.01134)				
Oracle Distortion			-0.17857*** (0.01402)	-0.1773*** (0.01895)	-0.19251*** (0.01688)	-0.19251*** (0.01689)
Median Gas Price				-0.58534*** (0.0251)	-0.65256*** (0.02657)	-0.65256*** (0.02646)
Price Move (t-25)					0.01059* (0.00555)	0.01059* (0.00555)
30-Day Volatility					-0.61038*** (0.06557)	-0.61038*** (0.06569)
Stablecoin Collateral					-2.86473*** (0.57637)	-2.86472*** (0.57647)
Stable Rate Debt						0.00049 (0.11593)
MEV Boost (t+1)	1.2398*** (0.12397)	1.20539*** (0.1263)	1.26232*** (0.12584)	1.35712*** (0.13756)	1.37317*** (0.14357)	1.37318*** (0.14355)
Intercept	-24.86398*** (0.58105)	-22.67374*** (0.51507)	-23.69697*** (0.59398)	-28.82691*** (0.80444)	-27.60724*** (1.05225)	-27.60735*** (1.05161)
$R^2_{McFadden}$	0.271	0.26	0.276	0.326	0.351	0.351
R^2_{Tjur}	0.089	0.084	0.092	0.145	0.157	0.157

*p<0.1; **p<0.05; ***p<0.01

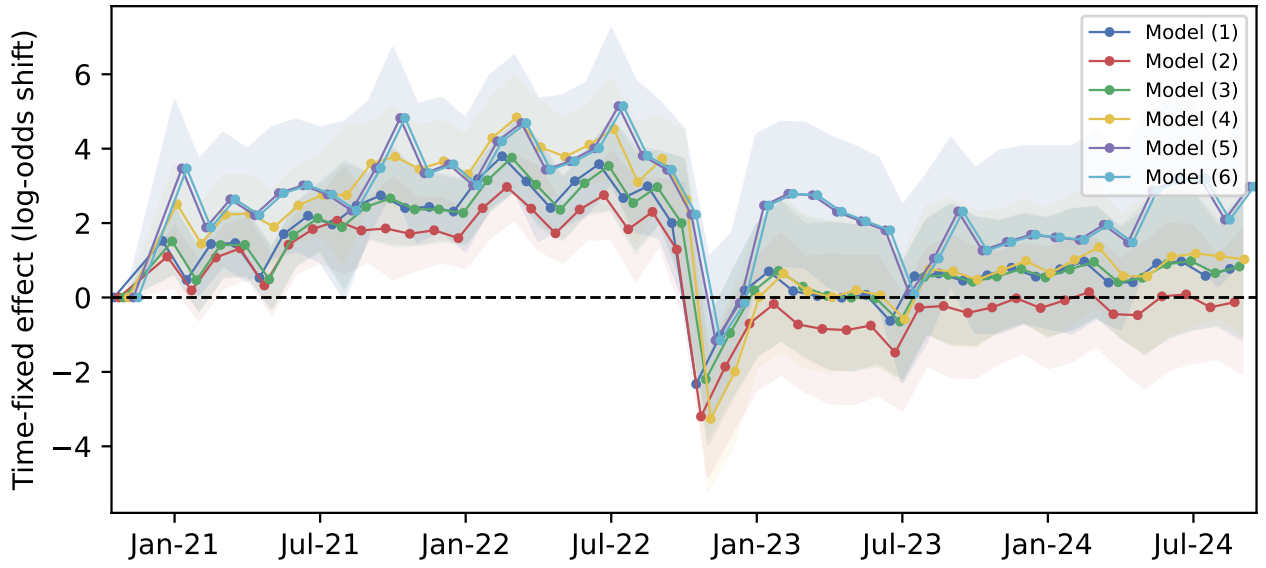


Figure 11: Estimated monthly time-fixed effects with 95% confidence intervals for each model specification estimated on observations using an alternative staleness cut-off of $\theta = 500$ finney. December 2020 serves as baseline month. Note: Lines for different model specifications are slightly horizontally offset to improve readability. The offset has no analytical meaning.

D.2 Interest Rate Model

Table 5: Logistic regression estimates of next-block liquidation probability for viable variable interest rate liquidation opportunities in the Aave V2 Main Market on Ethereum. This table reports coefficient estimates for the probability of liquidation in the next block $t + 1$ for variable rate borrower positions that satisfy $\bar{v}^{(k)} \geq \theta$ at block t . The dependent variable equals one if the position is liquidated at least once in the subsequent block and zero otherwise. The estimation sample includes all liquidation cases where $\bar{v}^{(k)} \geq 250$ fin at t . Controls are undersampled from the population of all borrower positions in state (V) at t that are not liquidated during $t + 1$, maintaining a 1:20 case-control ratio within monthly bins. All specifications include monthly time-fixed effects and intercepts are corrected for control undersampling following King and Zeng (2001). Cluster-robust standard errors are reported in parentheses.

Logistic Regression with Time-fixed Effects for Probability of Liquidation in $t + 1$ Reference = No Liquidation. 404 754 observations with $\bar{v}^{(k)} \geq 250$ fin covering 19 274 cases with controls randomly sampled out of 133 089 531 for a 1:20 case-control ratio in each month (46 bins).					
	(1)	(2)	(3)	(4)	(5)
Log Liquidation Volume	4.40582*** (0.07996)	4.35798*** (0.07948)	4.36955*** (0.07995)	5.36036*** (0.08313)	5.69516*** (0.08685)
Liquidator Bonus	0.00301*** (0.00023)		0.00309*** (0.00022)	0.00353*** (0.00027)	0.00386*** (0.00025)
Adjusted Bonus		0.06565*** (0.01284)			
Oracle Distortion			-0.0918*** (0.01399)	-0.07525*** (0.01577)	-0.08848*** (0.01481)
Median Gas Price				-0.48793*** (0.00056)	-0.59989*** (0.00067)
Price Move (t-25)					0.01064*** (0.00402)
30-Day Volatility					-0.6367*** (0.05761)
Stablecoin Collateral					-2.79755*** (0.60503)
MEV Boost (t+1)	1.61159*** (0.17076)	1.56992*** (0.17162)	1.62919*** (0.17214)	1.74548*** (0.19258)	1.87281*** (0.18353)
Intercept	-26.74653*** (0.54045)	-24.66418*** (0.55391)	-26.13444*** (0.55352)	-29.74694*** (0.62134)	-29.30691*** (0.95371)
$R^2_{McFadden}$	0.292	0.281	0.294	0.323	0.355
R^2_{Tjur}	0.088	0.082	0.09	0.121	0.143

*p<0.1; **p<0.05; ***p<0.01

Table 6: Logistic regression estimates of next-block liquidation probability for viable stable interest rate liquidation opportunities in the Aave V2 Main Market on Ethereum. This table reports coefficient estimates for the probability of liquidation in the next block $t + 1$ for stable rate borrower positions that satisfy $\bar{v}^{(k)} \geq \theta$ at block t . The dependent variable equals one if the position is liquidated at least once in the subsequent block and zero otherwise. The estimation sample includes all liquidation cases where $\bar{v}^{(k)} \geq 250$ fin at t . Controls are undersampled from the population of all borrower positions in state (V) at t that are not liquidated during $t + 1$, maintaining a 1:20 case-control ratio within monthly bins. All specifications include monthly time-fixed effects and intercepts are corrected for control undersampling following King and Zeng (2001). Cluster-robust standard errors are reported in parentheses.

Logistic Regression with Time-fixed Effects for Probability of Liquidation in $t + 1$ Reference = No Liquidation. 76 272 observations with $\bar{v}^{(k)} \geq 250$ fin covering 3632 cases with controls randomly sampled out of 32 113 551 for a 1:20 case-control ratio in each month (41 bins).					
	(1)	(2)	(3)	(4)	(5)
Log Liquidation Volume	3.94836*** (0.12461)	3.89954*** (0.12375)	3.89037*** (0.12447)	5.51664*** (0.17009)	5.518*** (0.17093)
Liquidator Bonus	0.00187*** (0.00021)		0.00184*** (0.00021)	0.00279*** (0.00024)	0.00274*** (0.00027)
Adjusted Bonus		0.05102*** (0.01826)			
Oracle Distortion			-0.15446*** (0.02472)	-0.14103*** (0.03103)	-0.13898*** (0.03086)
Median Gas Price				-0.57114*** (0.03193)	-0.57142*** (0.03205)
Price Move (t-25)					0.00213 (0.00316)
30-Day Volatility					0.00509 (0.05598)
Stablecoin Collateral					0.00097 (0.21501)
MEV Boost (t+1)	0.69137** (0.29589)	0.66497** (0.31429)	0.80663*** (0.28689)	0.94121*** (0.27083)	0.93899*** (0.27137)
Intercept	-22.54764*** (0.62121)	-21.3955*** (0.52674)	-21.43462*** (0.64607)	-27.02768*** (1.07978)	-27.02319*** (1.0874)
$R^2_{McFadden}$	0.217	0.21	0.221	0.274	0.274
R^2_{Tjur}	0.039	0.035	0.041	0.078	0.078

*p<0.1; **p<0.05; ***p<0.01

D.3 Ethereum Merge

Table 7: Logistic regression estimates of next-block liquidation probability for viable liquidation opportunities in the Aave V2 Main Market on pre-Merge Ethereum. This table reports coefficient estimates for the probability of liquidation in the next block $t + 1$ for borrower positions that satisfy $\bar{v}^{(k)} \geq \theta$ at block t . The dependent variable equals one if the position is liquidated at least once in the subsequent block and zero otherwise. The estimation sample includes all liquidation cases where $\bar{v}^{(k)} \geq 250$ fin at t . Controls are undersampled from the population of all borrower positions in state (V) at t that are not liquidated during $t + 1$, maintaining a 1:20 case-control ratio within monthly bins. All specifications include monthly time-fixed effects and intercepts are corrected for control undersampling following King and Zeng (2001). Cluster-robust standard errors are reported in parentheses.

Logistic Regression with Time-fixed Effects for Probability of Liquidation in $t + 1$ Reference = No Liquidation. 412 188 observations with $\bar{v}^{(k)} \geq 250$ fin covering 19 628 cases with controls randomly sampled out of 137 765 455 for a 1:20 case-control ratio in each month (22 bins).						
	(1)	(2)	(3)	(4)	(5)	(6)
Log Liquidation Volume	4.75227*** (0.06993)	4.70846*** (0.0702)	4.69002*** (0.07088)	6.37896*** (0.0842)	6.32384*** (0.08449)	6.31824*** (0.08462)
Liquidator Bonus	0.00233*** (0.00015)		0.00238*** (0.00015)	0.00314*** (0.0002)	0.00328*** (0.0002)	0.00328*** (0.0002)
Adjusted Bonus		0.04505*** (0.00977)				
Oracle Distortion			-0.14908*** (0.01373)	-0.12523*** (0.01476)	-0.12638*** (0.01447)	-0.12667*** (0.01441)
Median Gas Price				-0.69178*** (0.00056)	-0.69949*** (0.00065)	-0.69901*** (0.00067)
Price Move (t-25)					0.00578* (0.00299)	0.00581* (0.00305)
30-Day Volatility					-0.22611*** (0.03656)	-0.22513*** (0.03674)
Stablecoin Collateral					-0.77512*** (0.18355)	-0.78045*** (0.1848)
Stable Rate Debt						-0.08968 (0.08905)
Intercept	-26.97908*** (0.60392)	-25.28531*** (0.55759)	-25.89916*** (0.61796)	-32.06644*** (0.76316)	-31.24923*** (0.87451)	-31.21212*** (0.87351)
$R^2_{McFadden}$	0.308	0.3	0.312	0.364	0.365	0.365
R^2_{Tjur}	0.093	0.087	0.096	0.148	0.148	0.148

*p<0.1; **p<0.05; ***p<0.01

Table 8: Logistic regression estimates of next-block liquidation probability for viable liquidation opportunities in the Aave V2 Main Market on post-Merge Ethereum. This table reports coefficient estimates for the probability of liquidation in the next block $t + 1$ for borrower positions that satisfy $\bar{v}^{(k)} \geq \theta$ at block t . The dependent variable equals one if the position is liquidated at least once in the subsequent block and zero otherwise. The estimation sample includes all liquidation cases where $\bar{v}^{(k)} \geq 250$ fin at t . Controls are undersampled from the population of all borrower positions in state (V) at t that are not liquidated during $t + 1$, maintaining a 1:20 case-control ratio within monthly bins. All specifications include monthly time-fixed effects and intercepts are corrected for control undersampling following King and Zeng (2001). Cluster-robust standard errors are reported in parentheses.

Logistic Regression with Time-fixed Effects for Probability of Liquidation in $t + 1$ Reference = No Liquidation. 68 838 observations with $\bar{v}^{(k)} \geq 250$ fin covering 3278 cases with controls randomly sampled out of 27 437 627 for a 1:20 case-control ratio in each month (25 bins).						
	(1)	(2)	(3)	(4)	(5)	(6)
Log Liquidation Volume	2.7439*** (0.20316)	2.37751*** (0.25941)	2.77606*** (0.19783)	2.72789*** (0.23514)	3.96094*** (0.25158)	3.97667*** (0.25346)
Liquidator Bonus	0.01217*** (0.00189)		0.01304*** (0.00189)	0.01277*** (0.00159)	0.01237*** (0.00138)	0.01211*** (0.00131)
Adjusted Bonus		0.12293*** (0.03443)				
Oracle Distortion			-0.06821*** (0.0197)	-0.06885*** (0.01889)	-0.10768*** (0.02293)	-0.10645*** (0.02258)
Median Gas Price				0.10749 (0.13412)	-0.50488*** (0.1112)	-0.50313*** (0.11176)
Price Move (t-25)					0.02811** (0.01329)	0.02258* (0.01197)
30-Day Volatility					-2.36782*** (0.23729)	-2.34858*** (0.23314)
Stablecoin Collateral					-1.61408*** (0.53251)	-1.64887*** (0.53552)
Stable Rate Debt						0.74752 (0.76438)
Intercept	-23.61543*** (1.25905)	-16.08514*** (0.75216)	-23.78867*** (1.27075)	-23.50406*** (1.08299)	-22.22469*** (1.09115)	-22.18573*** (1.0772)
$R^2_{McFadden}$	0.23	0.162	0.235	0.234	0.356	0.356
R^2_{Tjur}	0.056	0.029	0.059	0.057	0.139	0.135

*p<0.1; **p<0.05; ***p<0.01